

GUNA THIAGARAJAN

Research Software Engineer | Python, Machine Learning & Large Sensor Datasets

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PROFILE

I write scientific software that has to run on real data. At **Fraunhofer UMSICHT** I built the full data path for CO2 electrolysis test benches. A Python service reads live sensor streams from many channels, repairs them and lines them up in time, then stores them in DuckDB for fast in-memory querying. Statistical detection and Random Forest forecasting sit on top. Four peer-reviewed publications. M.Sc. Sustainable Energy Systems, TU Dortmund, October 2026.

CORE SKILLS

Machine Learning & Data	Python (NumPy, Pandas, SciPy, scikit-learn , Matplotlib), unsupervised anomaly detection, statistical change-point detection, variance estimators, Random Forest, Isolation Forest, DBSCAN, time-series forecasting, feature engineering, SQL, DuckDB
Edge & Embedded Systems	Embedded C , interrupt service routines, Direct Memory Access, hardware FIFOs, I2C bus tuning, real-time clocks (DS3231), analog front ends (MAX86150), HX711 load-cell ADCs, H-bridge and PWM motor drivers, battery management, multi-domain voltage isolation
Test & Validation	Hardware-in-the-loop (HiL) testing, 8D root-cause analysis, ECU & infotainment diagnostics, test-bench instrumentation, ISO/IATF 16949
Industrial & Networks	OPC UA , Modbus, PROFINET, I2C, SPI, UART, PLC, digital twin architecture, LabVIEW
Tools & Environment	Git, Linux, Jupyter, VS Code, PyQt6, MATLAB/Simulink, COMSOL Multiphysics, LaTeX

PROFESSIONAL EXPERIENCE

Data Scientist / Automation Engineer Feb 2025 to Present

Fraunhofer UMSICHT · Oberhausen, Germany · Working student and Master's thesis

Built the data pipeline for a Cognitive Digital Twin of CO2 electrolysis test benches. A Python service (NumPy, Pandas, SciPy) reads the rig's logs while it is still running. It repairs broken rows and time-aligns every sensor channel into a local DuckDB for fast in-memory querying at the edge. A finished campaign is ready to analyse in about five minutes, down from hours.

Wrote the detection layer in scikit-learn over multi-channel electrolysis data. Variance estimators and statistical change-point detection watch the incoming distribution for structural breaks. A break usually means cell degradation or environmental drift. Random Forest models then forecast and correct against past behaviour to hold the stand at efficiency.

Turned raw time-series logs into Faradaic efficiency and energy-cost KPIs, computed live on the test stand rather than offline afterwards, so engineers read performance while the experiment is still running.

Replaced hand-written reporting for an intersectoral R&D team with a PyQt6 dashboard that writes its own reports. KPI and diagnostic results are now there when engineers want them, not when someone has time to make them.

Hardware & Validation Engineer Apr 2023 to Sep 2024

HARMAN International (Samsung Electronics) · Karlsbad, Germany · Working student, Hardware Analysis & Problem Solving team

Took the open warranty case count on the BMW and Mercedes-Benz Hermes infotainment programme from 123 into single digits within six months. Used 8D root-cause analysis and hardware-in-the-loop validation rather than fixing cases one at a time. HARMAN recognised the result in November 2023.

Wrote reusable HiL test procedures and Python scripts that parse diagnostic trace logs. Fault isolation that used to be ad hoc became a documented process the rest of the team could follow.

Checked embedded software and hardware against ISO/IATF 16949 on the BMW and Mercedes-Benz programmes. Wrote the analysis documentation that closed each defect to automotive audit standards.

Associate Software Engineer Jan 2022 to Aug 2022

Ernst & Young (EY) · Pune, India · Full-time

Replaced about five hours of manual reconciliation per reporting cycle with SQL and Python validation routines. Shipped iterative releases of a financial tracking application under audit-grade change control.

Analytics Associate Jun 2021 to Dec 2021

Verticurl (WPP Group) · Coimbatore, India · Full-time

Found the drop-off points behind conversion loss on international client accounts by segmenting large clickstream datasets in Google and Adobe Analytics and modelling user behaviour against funnel stages. Named top individual contributor with the Spotlight Award in 2021.

EDUCATION

M.Sc. Sustainable Energy Systems Oct 2024 to Oct 2026

TU Dortmund · Germany

Focus: Electromagnetic Field Theory Simulation (1.7), Selected Chapters in High Voltage Technology (1.7), Planning & Operation of Distributed Energy Sources (2.0), Smart Grids, Local Networks - Communication & Control.

Thesis: Cognitive Digital Twin for real-time monitoring and diagnostics of CO2 electrolysis cells. Built in Python, from raw sensor ingestion through KPI extraction and unsupervised anomaly detection to forecasting and a PyQt6 reporting front end.

German language course and university admission Sep 2022 to Mar 2023

Relocation from India to Germany · Germany

B.E. Electronics & Communication Engineering 2017 to 2021

Anna University / Sri Ramakrishna Institute of Technology · India

First Class, CGPA 8.29/10.

PUBLICATIONS, AWARDS & ADDITIONAL INFORMATION

Publications	SMART Bed for Elderly Care , Indian Journal of Public Health (Scopus), 2019. Plus 3 further undergraduate publications.
Awards	Single-Digit Backlog recognition , Hermes programme , HARMAN International, 2023 · Spotlight Award , Verticurl (WPP Group), 2021 · Grand Award , Asia Invention Association, 2019 · I2CreaTE Gold Award , Malaysia International Engineering Expo, 2019
Languages	English C1 · German A2, B1 course in progress · Tamil / Telugu Native · Hindi B2
Driving licence	EU Class A & B
Available from	1 November 2026